

Exercise 1:

$$(D_1): [1, 0, 1, 1, 0]$$

$$(D_2): [0, 1, 1, 1, 0]$$

$$(D_3): [1, 0, 0, 1, 1]$$

Exercise 2:

$$t_f(\text{cat}, D_1) = \frac{c(t, d)}{\sum_c (t, d)} = \frac{1}{3}$$

$$t_f(\text{eats}, D_1) = \frac{1}{3}$$

$$t_f(\text{fish}, D_1) = \frac{1}{3}$$

Exercise 3:

$$id_f(\text{cat}) = \log \frac{N}{d_f(\text{cat})} = \log \left(\frac{3}{2} \right)$$

$$id_f(\text{dog}) = \log \frac{N}{d_f(\text{dog})} = \log \left(\frac{3}{1} \right)$$

$$id_f(\text{eats}) = \log \left(\frac{3}{2} \right)$$

$$id_f(\text{fish}) = \log \left(\frac{3}{3} \right) = 0 \quad \begin{array}{l} \text{độ hiếm, nhiên } \rightarrow 3 \text{ docs} \\ \rightarrow \text{độ hiếm} = 0 \end{array}$$

$$id_f(\text{likes}) = \log \left(\frac{3}{1} \right)$$

Exercise 4:

$$\begin{aligned} t_{fid}(\text{cat}, D_1) &= t_f(\text{cat}, D_1) \times id_f(\text{cat}) \\ &= \frac{1}{3} \times \log \left(\frac{3}{2} \right) \end{aligned}$$

$$\begin{aligned} t_{fid}(\text{eats}, D_1) &= t_f(\text{eats}, D_1) \times id_f(\text{eats}) \\ &= \frac{1}{3} \times \log \left(\frac{3}{2} \right) \end{aligned}$$

$$\begin{aligned}
 \text{Jinfo}(\text{gish}, D) &= \text{Jg}(\text{gish}, D_1) \times \text{idg}(\text{gish}) \\
 &= \frac{1}{3} \times \log\left(\frac{3}{3}\right) = 0
 \end{aligned}$$

Exercise 5:

$$x = [1, 1, 1]$$

$$y = [1, 1, 0]$$

$$\cos(x, y) = \frac{x^T y}{\|x\|_2 \|y\|_2}$$